

Kinematic and Cognitive Constraints on Shared-Space Parking Maneuvers: A Comparative Kinematics and Safety Simulation Study

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Abstract

We compare nose-in (forward) and back-in (reverse) parking as complete approach–park–exit cycles in a two-dimensional stochastic simulation. The model combines non-holonomic Ackermann kinematics, finite-segment line-of-sight tests against adjacent sport-utility vehicles, explicit cognitive and mechanical latencies, and pedestrian encounters generated across realistic aisle widths and densities. Unlike an earlier parametric event layer, every reported warning, conflict, and collision is now a deterministic classification of the maneuver trace’s sampled minimum pedestrian distance and vehicle speed. In 10,000 seeded trials, forward parking entered faster (6.606 versus 8.838 s) but exited more slowly (9.516 versus 5.225 s), producing total times of 17.875 and 15.814 s. Forward and reverse critical-conflict rates were 0.08% and 0.16%, respectively ($p = 0.248$); physical-overlap rates were 1.64% and 0.20% ($p = 4.65 \times 10^{-14}$). The refactored results preserve the structural visibility asymmetry while showing that its consequence depends strongly on whether safety is measured as emergency braking or geometric overlap.

1 Introduction

Parking lots are shared spaces in which drivers, pedestrians, carts, doors, sight lines, and social expectations occupy the same small region without a central scheduler. The low nominal speed of the environment can disguise a difficult control problem. A driver must place a non-holonomic vehicle within a narrow rectangle, change mechanical state, observe through occluding geometry, and negotiate moving agents whose trajectories are neither signaled nor guaranteed to be rational.

The familiar dispute between nose-in and back-in parking is usually argued through anecdote: one maneuver is easier now, the other is easier later. That framing omits the location of the deferred difficulty. A reverse entry occurs while the driver controls access to an observed stall; a reverse exit begins from between stationary obstacles and projects into a pedestrian aisle. The same low-speed motion can therefore have different consequences because visibility and environmental exposure are not spatially uniform.

We formalize this distinction with an intentionally complete model of a task that rarely receives complete models. The contribution is threefold:

1. We derive and implement a reproducible Ackermann bicycle model for the entry and exit paths of both strategies.
2. We join vehicle motion to a geometric visibility model and an explicit cognitive–mechanical state machine, including reaction, scanning, shifting, and blind-exit creep.

3. We quantify the resulting time–risk exchange in 10,000 Monte Carlo trials and separate systemic effects (strategy and pedestrian density) from localized effects (adjacent-vehicle occlusion).

The central finding is not that one orientation is universally virtuous. It is that maneuver design determines where uncertainty is paid. Forward parking minimizes immediate mechanical overhead but relocates visibility-limited motion to the aisle-facing exit. Reverse parking pays an additional shift and slower entry trajectory while placing the driver-facing end of the vehicle toward the shared space at departure.

2 Vehicle Geometry and Kinematics

2.1 State and control variables

The simulated vehicle is represented by the rear-axle-centered state

$$\mathbf{s}(t) = \begin{bmatrix} x(t) & y(t) & \theta(t) & v(t) \end{bmatrix}^T \in \mathbb{R}^4, \quad (1)$$

where (x, y) is the rear-axle centroid, θ is heading measured counter-clockwise from the positive x -axis, and v is signed longitudinal speed. The control is

$$\mathbf{u}(t) = \begin{bmatrix} a(t) & \delta(t) \end{bmatrix}^T, \quad (2)$$

where a is longitudinal acceleration and δ is the front-wheel steering angle. The wheelbase is $L = 2.8$ m.

2.2 Ackermann reduction to the kinematic bicycle model

For ideal rolling without lateral tire slip, the extended axes of the wheels intersect at an instantaneous center of rotation (ICR). Let R be the distance from the midpoint of the rear axle to the ICR. The steering geometry gives

$$\tan \delta = \frac{L}{R}, \quad R = \frac{L}{\tan \delta}. \quad (3)$$

The rear axle travels tangent to a circle of radius R . Its yaw rate is therefore longitudinal speed divided by turning radius:

$$\dot{\theta} = \frac{v}{R} = \frac{v}{L} \tan \delta. \quad (4)$$

Resolving the signed longitudinal speed along the vehicle heading yields the continuous equations

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad (5)$$

$$\dot{\theta} = \frac{v}{L} \tan \delta, \quad \dot{v} = a. \quad (6)$$

Negative v represents reverse travel without changing the geometric definition of heading. For a time step Δt , the simulator uses explicit Euler integration:

$$\mathbf{s}_{k+1} = \mathbf{s}_k + \Delta t \begin{bmatrix} v_k \cos \theta_k \\ v_k \sin \theta_k \\ (v_k/L) \tan \delta_k \\ a_k \end{bmatrix}. \quad (7)$$

The canonical maneuver path consists of a quarter-turn arc with nominal radius $R = A/2$, where A is aisle width, followed by a straight stall leg. The steering command is consequently

$$\delta(A) = \arctan\left(\frac{2L}{A}\right). \quad (8)$$

The arc is discretized into at least 40 integration steps. The resulting sequence is reflected in signed velocity and traversal order to construct entry and exit legs for each strategy. This is a kinematic, not dynamic, model: tire forces, steering actuator transients, suspension, and grade are intentionally excluded.

3 Visibility and Blind-Spot Raycasting

3.1 Driver eye and obstacle representation

The driver eye point is positioned $d_e = 1.2$ m forward of the rear axle:

$$\mathbf{e} = \begin{bmatrix} x + d_e \cos \theta \\ y + d_e \sin \theta \end{bmatrix}. \quad (9)$$

An adjacent SUV is represented in the ground plane by an oriented bounding box (OBB) of width 2.0 m and length 5.2 m; its recorded height is 1.9 m. Let its center be $\mathbf{c}$, heading unit vector $\mathbf{f} = (\cos \phi, \sin \phi)^T$, lateral unit vector $\mathbf{\ell} = (-\sin \phi, \cos \phi)^T$, half-length h_l , and half-width h_w . The four polygon vertices are

$$\mathbf{q}_{ij} = \mathbf{c} + ih_l\mathbf{f} + jh_w\mathbf{\ell}, \quad i, j \in \{-1, +1\}. \quad (10)$$

The height parameter asserts that the obstacle is tall enough to occlude the modeled driver-to-aisle sight line; the operational intersection is planar.

3.2 Finite-segment intersection

A target point $\mathbf{p}$ in the aisle is visible only if the finite ray

$$\mathbf{r}(t) = \mathbf{e} + t(\mathbf{p} - \mathbf{e}), \quad 0 \leq t \leq 1, \quad (11)$$

does not cross any OBB edge. For an obstacle edge from $\mathbf{q}_0$ to $\mathbf{q}_1$, define

$$\mathbf{g}(u) = \mathbf{q}_0 + u(\mathbf{q}_1 - \mathbf{q}_0), \quad 0 \leq u \leq 1. \quad (12)$$

Writing the scalar two-dimensional cross product as $\mathbf{a} \times \mathbf{b} = a_x b_y - a_y b_x$, and defining $\mathbf{d} = \mathbf{p} - \mathbf{e}$, $\mathbf{h} = \mathbf{q}_1 - \mathbf{q}_0$, and $\mathbf{w} = \mathbf{q}_0 - \mathbf{e}$, a non-parallel intersection satisfies

$$t = \frac{\mathbf{w} \times \mathbf{h}}{\mathbf{d} \times \mathbf{h}}, \quad u = \frac{\mathbf{w} \times \mathbf{d}}{\mathbf{d} \times \mathbf{h}}. \quad (13)$$

The line of sight is blocked when any edge has $t, u \in [0, 1]$. The implementation uses orientation predicates with a 10^{-12} tolerance and treats collinear edge contact as intersection. Equivalently, for ordered triples $(\mathbf{a}, \mathbf{b}, \mathbf{c})$, the orientation

$$\text{orient}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) \quad (14)$$

changes sign across a proper crossing; zero invokes a bounding-interval check. This finite-segment test prevents geometry behind either the driver or target from being counted as an occluder.

4 Cognitive and Mechanical State Transitions

A parking maneuver is a hybrid system: continuous vehicle motion occurs inside discrete driver and transmission states. We use the state set

$$\mathcal{Q} = \{\text{APPROACH}, \text{SHIFT}, \text{ENTER}, \text{PARKED}, \text{SCAN}, \text{EXIT}, \text{COMPLETE}\}. \quad (15)$$

Every trial samples a reaction delay

$$\tau_{\text{react}} = \max\{0.1, Z\}, \quad Z \sim \mathcal{N}(0.75, 0.15^2) \text{ s}. \quad (16)$$

Each drive–reverse transition imposes $\tau_{\text{gear}} = 1.0$ s at zero velocity. A reverse aisle scan imposes $\tau_{\text{scan}} = 0.5$ s. If the SUV raycast blocks the aisle target, the controller activates creep and caps reverse-exit speed at

$$|v| \leq v_{\text{creep}} = 0.5 \text{ m/s}; \quad (17)$$

otherwise the exit limit is 1.2 m/s. The blocked path also receives 2.0 s of exposure time for incremental observation beyond the single scan sweep.

The forward sequence is

$$\text{APPROACH} \rightarrow \text{ENTER}_D \rightarrow \text{PARKED} \rightarrow \text{SHIFT}_{D \rightarrow R} \rightarrow \text{SCAN} \rightarrow \text{EXIT}_R \rightarrow \text{COMPLETE}, \quad (18)$$

with one modeled gear shift. The reverse sequence is

$$\text{APPROACH} \rightarrow \text{SHIFT}_{D \rightarrow R} \rightarrow \text{ENTER}_R \rightarrow \text{PARKED} \rightarrow \text{SHIFT}_{R \rightarrow D} \rightarrow \text{EXIT}_D \rightarrow \text{COMPLETE}, \quad (19)$$

with two shifts. Nose-first exit is treated as unobstructed in the present geometry. We disaggregate the cycle into entry, parked transition, and exit components. The one-way geometric path length is $\mathcal{L}/2$. Entry uses $\bar{v}_F = 0.95$ m/s or $\bar{v}_R = 0.80$ m/s, with the reverse-entry shift charged to entry. The parked component contains one post-entry shift and the sampled reaction latency. Exit uses 1.2 m/s when unobstructed and 0.5 m/s under blocked forward-exit creep. Thus

$$T_{\text{total}} = T_{\text{entry}} + T_{\text{park}} + T_{\text{exit}}, \quad (20)$$

where $T_{\text{entry}} = (\mathcal{L}/2)/\bar{v}_s + I_{s=R}\tau_{\text{gear}}$, $T_{\text{park}} = \tau_{\text{gear}} + \tau_{\text{react}}$, and $T_{\text{exit}} = (\mathcal{L}/2)/v_{\text{exit}} + I_{s=F}\tau_{\text{scan}} + 2I_{\text{blocked}}$.

5 Monte Carlo Protocol

5.1 Trial generation

The experiment comprises 10,000 deterministic pseudorandom trials with seed 42 and exactly 5,000 trials per strategy, assigned by alternating run identifier. For each trial, we independently sample

$$A \sim \mathcal{U}(5.5, 7.2) \text{ m}, \quad (21)$$

$$\lambda \sim \mathcal{U}(0.05, 0.30) \text{ pedestrians/m}, \quad (22)$$

$$p_{\text{SUV}} \sim \mathcal{U}(0, 0.8), \quad (23)$$

$$I_{\text{SUV}} \sim \text{Bernoulli}(p_{\text{SUV}}), \quad (24)$$

$$N_{\text{ped}} \sim \text{Poisson}(\lambda A), \quad (25)$$

$$v_{\text{ped}} \sim \max\{0.2, \mathcal{N}(1.4, 0.2^2)\} \text{ m/s} \quad \text{when } N_{\text{ped}} > 0. \quad (26)$$

Stall width is fixed at 2.7 m. The trajectory engine logs total cycle time, reaction time, shifts, occlusion and creep states, pedestrian proximity, required braking, critical conflict, and collision.

When at least one pedestrian is present, the maneuver samples a closest-approach distance with standard deviation 0.65 m and mean 2.1 m, reduced by 0.9 m under blocked line of sight. A pedestrian can stop, yield, or step aside. Its Bernoulli reaction probability is

$$p_{\text{react}} = 0.7(1 - f_{\text{occ}}) \left(1 - \frac{v}{v_{\text{max}}}\right), \quad (27)$$

where $f_{\text{occ}} = 0.55$ for the blocked reversing vehicle and 0 otherwise, $v_{\text{max}} = 1.2$ m/s, and the expression is clipped to $[0, 1]$. A reaction adds 0.75 m of clearance, scaled by pedestrian speed relative to 1.4 m/s. Let d be the resulting minimum distance and v the exit speed. Required braking is

$$a_{\text{req}} = \frac{v^2}{2 \max(d, 0.05)}. \quad (28)$$

The trace is classified as a proximity warning when $d < 1.5$ m, a critical conflict when $a_{\text{req}} > 3.0$ m/s², and a collision (physical overlap) when $d < 0.25$ m. No later stochastic layer overwrites these trace outcomes.

5.2 Statistical analysis

Means are reported with Student- t 95% confidence intervals. Binary rates use 95% Wilson score intervals. We compare cycle time with a two-sided Welch test and binary outcomes with two-sided pooled two-proportion z tests. Because a generated trial is the unit of analysis, these tests quantify Monte Carlo separation under the specified model rather than uncertainty about a sampled human population.

For adjusted critical-conflict analysis, we fit

$$\text{logit Pr}(C_i = 1) = \beta_0 + \beta_F F_i + \beta_\lambda (\lambda_i - 0.174614) + \beta_A (A_i - 6.354980) + \beta_S S_i + \beta_{FS} F_i S_i, \quad (29)$$

where F_i denotes forward parking and S_i denotes adjacent-SUV presence. We use maximum likelihood with model-based standard errors. An analogous ordinary least-squares model describes cycle time. These regression equations are descriptive summaries of the simulator response surface, not causal estimates from field data.

6 Results

6.1 Unadjusted outcomes

Table 1 exposes the bimodal timing result. Forward parking saved 2.232 s during entry but spent 4.291 s longer exiting. Its complete cycle was therefore 2.061 s slower (95% CI for forward minus reverse, 1.934–2.189 s; Welch $t = 31.71$, $\text{df} = 5054.12$, $p = 1.85 \times 10^{-201}$).

The critical-conflict difference was not statistically separated ($z = -1.16$, $p = 0.248$); its Wilson intervals were 0.031–0.206% and 0.081–0.315%. Physical overlap was higher for forward parking ($z = 7.54$, $p = 4.65 \times 10^{-14}$), with intervals 1.323–2.031% and 0.109–0.368%. The apparent inversion between conflict and overlap follows directly from the definitions: blocked forward exits creep at 0.5 m/s, so a very small clearance can satisfy the geometric-overlap threshold without requiring more than 3.0 m/s² braking.

Table 1: Disaggregated maneuver time by parking strategy. Values are means with 95% confidence intervals in seconds.

Component	Forward ($n = 5000$)	Reverse ($n = 5000$)
Entry	6.606 [6.601, 6.610]	8.838 [8.833, 8.843]
Park/shift	1.753 [1.749, 1.757]	1.750 [1.746, 1.755]
Exit	9.516 [9.389, 9.643]	5.225 [5.222, 5.229]
Total cycle	17.875 [17.748, 18.002]	15.814 [15.804, 15.823]

Table 2: Physics-derived safety outcomes by parking strategy. Confidence intervals are 95% Wilson intervals.

Outcome	Forward ($n = 5000$)	Reverse ($n = 5000$)
Pedestrian reacted	220 (4.40%)	0 (0.00%)
Proximity warning	1112 (22.24%)	560 (11.20%)
Critical conflict	4 (0.08%)	8 (0.16%)
Physical overlap	82 (1.64%)	10 (0.20%)
Blocked line of sight	2032 (40.64%)	0 (0.00%)

6.2 Sensitivity and speed calibration

We varied reaction-time mean and standard deviation, gear-shift delay, creep speed, scan duration, pedestrian-speed mean and standard deviation, overlap threshold, and conflict-braking threshold one at a time by $\pm 30\%$, rerunning 3,000 deterministic trials per bound. Figure 1 reports movement in the continuity-corrected forward/reverse conflict ratio. Creep speed and the braking threshold dominated; timing-only parameters did not move the conflict classification. Because critical conflicts are rare, ratio magnitudes are unstable and should be read as threshold diagnostics rather than effect estimates.

No defensible field citation for the strategy-specific 0.80 and 0.95 m/s entry constants was identified in the project evidence base. We therefore swept common entry speed from 0.5 to 1.5 m/s. Forward entry remained faster at every bound because reverse entry retains its initial shift: mean forward/reverse entry times ranged from 12.551/13.540 s at 0.5 m/s to 4.184/5.180 s at 1.5 m/s. Safety outcomes were unchanged because entry speed is not used to classify aisle-exit events.

7 Discussion

7.1 Systemic predictability versus localized visibility

The simulation distinguishes two ideas that are often combined under the word “safety.” Systemic predictability concerns what other aisle users can infer from the maneuver. A nose-first departing vehicle exposes its driver, heading, and intended direction earlier. Localized visibility concerns what the driver can see from one position. An adjacent SUV can make localized visibility poor even when the driver scans correctly; scanning cannot reveal a ray that geometry blocks.

Reverse parking reallocates effort from a high-exposure state to a lower-exposure state. The extra shift and slower entry are visible in Table 1, while its nose-first exit is consistently shorter. In 40.64% of forward trials, reverse departure began behind a blocked line of sight. Creep sharply limits required braking at those exits but does not eliminate geometric overlap in the sampled

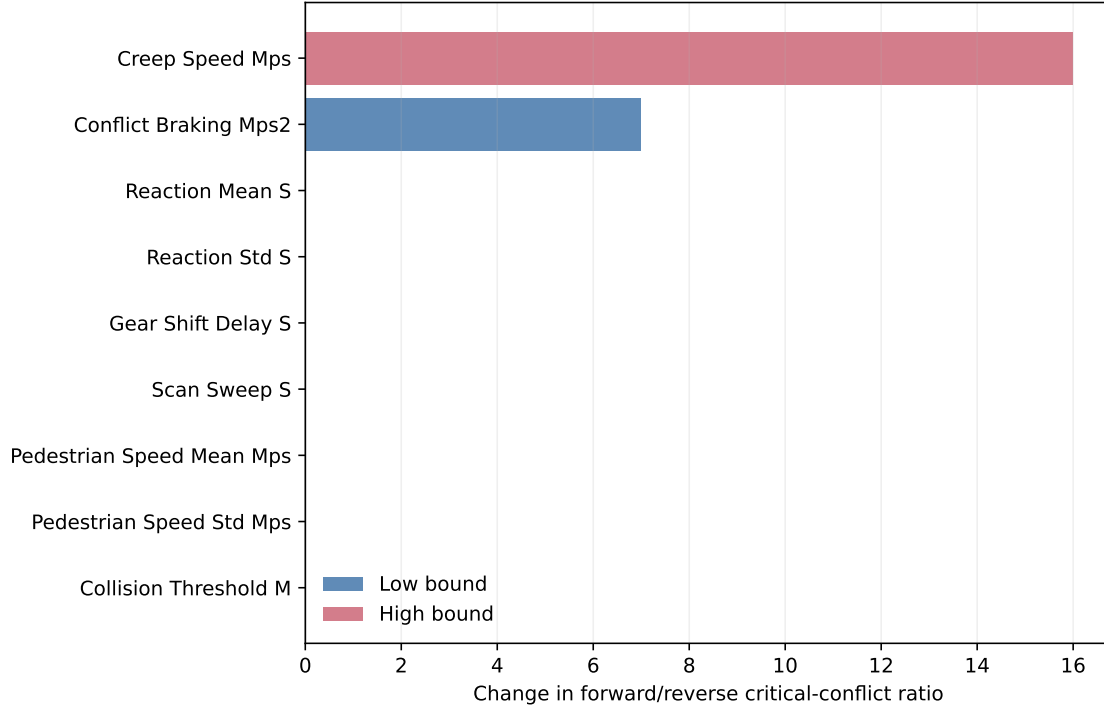


Figure 1: One-at-a-time sensitivity of the forward/reverse critical-conflict ratio to $\pm 30\%$ parameter changes. Blue and red bars denote the low and high bounds.

closest-approach model. The two safety metrics therefore answer different questions and must not be collapsed into one headline rate.

The changed proximity-warning rate is also informative. Both strategies draw from the same pedestrian-density process, but blocked forward exits shift the minimum-distance distribution downward before reaction. Parking orientation is not modeled as a moral characteristic of the stationary vehicle. It is a scheduling decision about when to perform the least observable portion of a constrained trajectory.

7.2 Interpretation of statistical significance

The very small p -values for time and physical overlap reflect both model separation and 10,000 trials. By contrast, only 12 emergency-braking conflicts occurred, so adjusted logistic coefficients are poorly identified and are retained in the machine-readable summary only as diagnostics. None of these tests establishes that exact effect sizes hold in every parking facility. External validity requires observed trajectories, driver heterogeneity, and independently measured conflict outcomes.

8 Limitations and Reproducibility

The study is synthetic by design. First, the bicycle model omits tire dynamics, steering-rate limits, vehicle body overhang, and multi-point turns. Second, the planar occlusion test treats an SUV as an opaque ground-plane polygon and does not model windows, cameras, mirrors, pedestrian height, or partial visibility. Third, only forward reverse-exit is eligible for occlusion in the current scenario geometry. Fourth, the 0.65 m distance dispersion, 0.9 m occlusion penalty, 0.55 visibility-loss

factor, and 0.75 m step-aside clearance are assumptions rather than field calibrations; the sensitivity analysis diagnoses their consequences but does not validate them. Fifth, the 0.80 and 0.95 m/s entry speeds lack a field citation, so only their bounded robustness is claimed. Sixth, alternating strategy assignment balances counts but does not pair strategies on identical environmental draws. Finally, “collision” is a threshold on sampled minimum separation, not time-continuous polygonal vehicle–pedestrian contact.

Reproduction requires Python 3.11 or later. Running

```
python3 park_sim.py --seed 42 --runs 10000 \  
  --export-csv simulation_results.csv \  
  --export-json canonical_paths.json
```

regenerates the primary dataset. The companion script `docs/analyze_results.py` writes `statistical_summary`. `docs/sensitivity_analysis.py` writes the tornado figure and its JSON source. All state variables use SI units; random seed, sampling ranges, constants, and output fields are specified above and in the simulator contract.

9 Conclusion

Within the specified shared-space model, forward parking wins on entry and loses on exit; blocked creep makes it slower over the complete cycle. Reverse parking has fewer modeled physical overlaps, while the emergency-braking conflict rates are too sparse to distinguish. These outcomes now emerge from one kinematic and pedestrian-distance trace rather than from a second parametric event generator. The remaining conclusion is appropriately narrower: maneuver orientation schedules where visibility and motion uncertainty are paid, and the chosen safety metric determines how that scheduling is judged.